

# PocketQuake

From a catalog CSV to a relocation notebook in one command

---

Min-Seong Seo / SGTL, SNU

# What is PocketQuake?

A thin orchestrator that takes a list of earthquakes and produces a relocation + focal-mechanism notebook. It glues two specialised submodules:

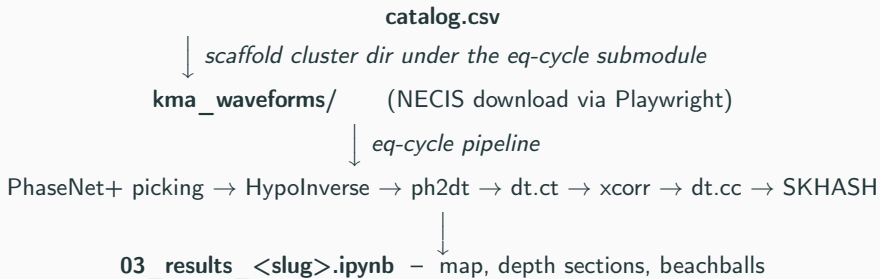
- **seismoseo/necis-downloader** — KMA NECIS waveform fetcher (Playwright + the FTP staging server).
- **seismoseo/korea-cluster-relocation** — the picking + HypoInverse + HypoDD + SKHASH pipeline.

**One command, end to end:**

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

Output: an executed Jupyter notebook with maps, depth sections, and beachball diagrams for every event you fed in.

# Workflow at a glance



You write the catalog and run one command. PocketQuake takes care of dispatching to the right waveform source, registering the cluster, and executing the notebook.

# Prerequisites

**OS** Linux or macOS (bash wrapper; Windows uses WSL)

**glibc**  $\geq 2.28$  — Ubuntu 20.04+, RHEL/Rocky 8+, Debian 10+. CentOS 7 / RHEL 7 are below the line (Playwright fails).

**Python** 3.10+ (pinned by `environment.yml`)

**conda** Miniforge / Miniconda / Mambaforge

**gcc/gfortran**  $\geq 9.3$  (for the compiled externals)

**Disk**  $\sim 10$  GB (conda env + Chromium + waveforms)

**Network** HTTPS to `github.com`, `code.usgs.gov`, `conda-forge`, `necis.kma.go.kr`

**NECIS account** (apply at `https://necis.kma.go.kr`, research / institutional accounts only; approval 1–5 days).

**STP account** is optional — only for pre-2020 waveforms.

## Install [1/2] — Python environment

```
# 1. Clone with submodules
git clone --recurse-submodules https://github.com/seismoSEO/PocketQuake.git
cd PocketQuake

# 2. Create the conda env (one command does everything Python-side)
conda env create -f environment.yml
conda activate pocketquake

# 3. Editable installs for PocketQuake itself + the NECIS submodule
pip install -e . -e external/necis-downloader

# 4. Playwright Chromium (one-time, ~200 MB)
playwright install chromium

# 5. Credentials
cp .env.example .env
# then edit NECIS_USER, NECIS_PASS
```

## Install [2/2] — External programs

PocketQuake's default chain calls these from the shell. None are pip-installable. Put each binary on \$PATH:

- hyp1.40 — HypoInverse (USGS Fortran)
- ph2dt + hypoDD — Waldhauser's HypoDD (Fortran)
- mseed2sac — IRIS converter (C)
- libarchive / bsdtar — already in environment.yml

Two research clones reachable by env var:

```
git clone https://github.com/AI4EPS/EQNet.git
echo "EQNET_DIR=$PWD/EQNet" >> ~/PocketQuake/.env

git clone https://code.usgs.gov/esc/SKHASH.git
echo "SKHASH_DIR=$PWD/SKHASH/SKHASH" >> ~/PocketQuake/.env
```

Full per-tool source + build instructions: **docs/EXTERNAL\_TOOLS.md**.

A plain CSV with 10 columns, KST origin times:

```
Year,Month,Day,Hour,Minute,Second,Latitude,Longitude,Magnitude,Depth
2025,2,7,2,35,34,37.14,127.76,3.1,9
2025,2,7,2,54,38,37.14,127.76,1.4,6
2025,2,7,3,49,4,37.14,127.76,1.5,7
2025,2,8,10,13,23,37.14,127.76,1.6,7
```

- Origin time interpreted as KST (UTC + 9 h)
- PocketQuake auto-derives the cluster epicenter (catalog centroid) and the spatial bounds (bbox + 0.2°); override with `-epi`, `-bounds`
- Magnitude is informational — relocation does not use it
- Depth  $\geq 0$ , measured in km

# Run the chungju example

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

The wrapper:

1. scaffolds a cluster dir under  
external/korea-cluster-relocation/chungju\_cluster/
2. downloads NECIS waveforms (Playwright drives the JS portal, polls the staging queue, streams the FTP files)
3. runs PhaseNet+ → HypoInverse → HypoDD → SKHASH
4. builds + executes the results notebook

Useful flags:

- -source mixed — per-event STP+NECIS dispatch
- -picker stead — SeisBench PhaseNet (no EQNet needed)
- -mainshock <UTC> — Gwangyang-style mainshock treatment
- -cores N — cap xcorr workers (default 10)



# What you see while it runs

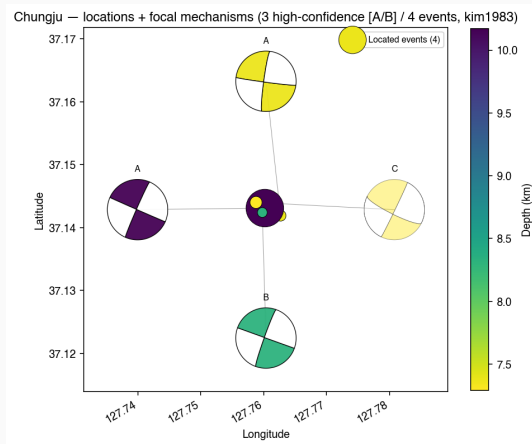
**Foreground** (-fg): full log to terminal.

**Default** (background): nohup.out in the working directory.

```
[pocketquake] scaffolding chungju_cluster ...  
[necis] login OK; submitting request for 4 event(s)  
[necis] request id 4f3a... status=P -> C, downloading 4 SAC bundles  
[eqcycle] picking with phasenet_plus (~30s/event)  
[eqcycle] hypoinverse: 4 events located  
[eqcycle] xcorr 6 pairs (10 workers) ...  
[eqcycle] dt.cc -> hypoDD reloc OK (4 events)  
[eqcycle] focal_mechanism: 4 SKHASH solutions  
[pocketquake] executing 03_results_chungju.ipynb
```

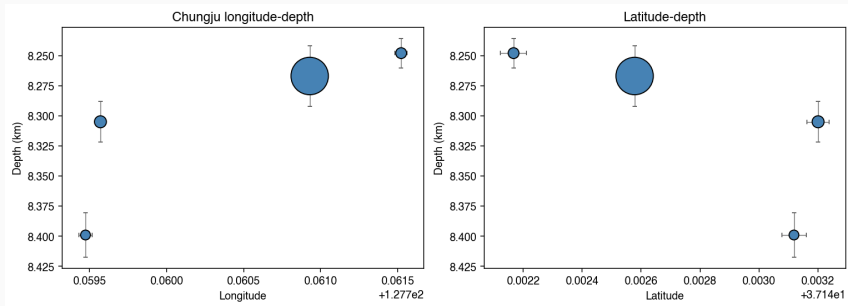
Wall-clock for chungju: ~15 min on a 16-core box.

## Output 1 — Relocated map + focal mechanisms



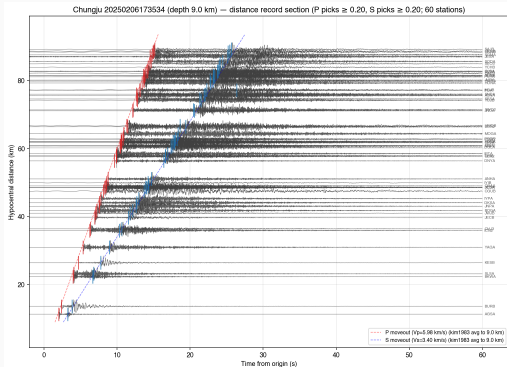
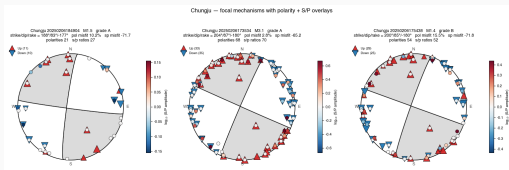
Four located events (depth-coloured dots) at ( $37.142^{\circ}\text{N}$ ,  $127.760^{\circ}\text{E}$ ) with quality A/B/C SKHASH beachballs drawn on a leader-line ring. All four solutions converge on a near-vertical N–S right-lateral strike-slip plane.

## Output 2 — Depth sections



Three orthogonal cross-sections (E–W, N–S, along-strike) showing the depth distribution. HypoDD relative-relocation tightens the cluster from  $\sim 1$  km scatter (KMA bulletin) to  $\sim 200$  m.

## Output 3 — Beachball gallery + record section



The notebook bundles a per-event beachball gallery (left) and a record-section view of the M3.1 mainshock with PhaseNet+ pick markers, first-motion polarities, and the SKHASH-inferred S/P amplitude ratios (right).

# Run on your own catalog

1. Save your KMA-style CSV (the format from slide 6).
2. One command:

```
./pocketquake.sh ~/my_catalog.csv myswarm
```

3. Override the auto-derived epicenter / bounds if your catalog spans a wide region:

```
./pocketquake.sh ~/my_catalog.csv myswarm \  
  --epi 35.46,128.43 \  
  --bounds 35.3,35.65,128.25,128.65
```

4. Mainshock treatment (re-runs xcorr→dt.cc with the mainshock excluded, builds a `_main` notebook):

```
./pocketquake.sh ~/my_catalog.csv myswarm \  
  --mainshock 20240912144719
```

## Common pitfalls (docs/TROUBLESHOOTING.md)

python interpreter not found set POCKETQUAKE\_PYTHON=/path/to/python in .env or put python3 on \$PATH.

Permission denied (publickey) use the HTTPS clone URL, not SSH.

Submodule path '...' not registered you forgot -recurse-submodules; fix with git submodule update -init -recursive.

EQNET\_DIR not set PhaseNet+ is the default picker; either install EQNet (slide 5) or pass -picker stead.

libstdc++ version not found glibc/libstdc++ too old (CentOS 7); upgrade OS or run the NECIS download elsewhere.

NECIS\_USER empty fill in .env.

## Where to go next

- **Read the executed result notebook** —  
`external/korea-cluster-relocation/pipeline/notebooks/03_results_<slug>.ipynb`
- **Inspect the auto-derived configuration** —  
`external/korea-cluster-relocation/pipeline/clusters/<slug>.py`
- **Tune the run** — `./pocketquake.sh -help`
- **Other examples bundled** — `examples/sangju/`, `examples/chungju/`, the cluster modules registered in `korea-cluster-relocation/pipeline/clusters/`
- **Docs** — `docs/INSTALL.md`, `docs/EXTERNAL_TOOLS.md`, `docs/TROUBLESHOOTING.md`

<https://github.com/seismoseo/PocketQuake>

*Made by Min-Seong Seo / SGTL, SNU.*